

Assignment V: Goal-Based Portfolio Optimisation

Objective

This assignment introduces goal-based investing, where the focus is on maximizing the probability of achieving financial goals rather than only optimizing returns. Students will implement a multi-goal portfolio allocation framework using Monte Carlo simulation.

Problem Setup

You are designing an investment plan for a client with the following goals:

Current Age of person: 25 years

Goal 1: Marriage — Target Amount: ₹25 L — Time : 4 years from now

Goal 2: Child Education — Target Amount: ₹30 L — Time : 12 years from now

Goal 3: House Purchase — Target Amount: ₹1Cr — Time Horizon: 15 years from now

Ultimate Goal: Retirement fund with SWP from 55-80

Inflation rate: 6% annually

Part 1: Goal Modeling and Future Value

Task 1.1: Inflation Adjustment

Convert each goal into its future value using inflation.

Rank goals based on urgency (shorter tenure implies higher priority).

Task 1.2: Minimum SIP Estimation

To reach each goal with 90% certainty, estimate the minimum SIP required to achieve the target using expected returns.

Use iterative or numerical solving methods.

Part 2: Monte Carlo Goal Probability

Task 2.1: Portfolio Simulation

Given the list of 5 securities. Create a portfolio of securities and generate the return paths using Monte Carlo simulation based on mean returns and covariance matrix.

Task 2.2: Goal Success Probability

Initial SIP budget: ₹20,000 per month

For each goal:

- Simulate $N = 5000$ paths
- Compute probability of success as the fraction of successful outcomes

Part 3: SIP Allocation Optimization

Task 3.1: Optimization Objective

Allocate SIP across goals to maximize total probability of success.

Constraints:

- Total SIP must equal ₹40,000

- Each goal must receive a minimum allocation

Task 3.2: Optimization Method

Use Grid Search or SLSQP optimization.

Output optimal SIP allocation and corresponding probabilities.

Part 4: Sensitivity Analysis

Task 4.1: Tenure vs Probability

For each goal:

- Increase tenure from current value up to +10 years (step size = 1 year)
- Keep SIP constant
- Plot tenure vs probability

Task 4.2: SIP vs Probability

For each goal:

- Increase SIP from +5% to +50% (step size = 5%)
- Plot SIP vs probability

Part 5: Portfolio Growth Visualization

Task 5.1: Growth Path

Plot:

- Median portfolio growth
- 10th and 90th percentile bands

Task 5.2: Dashboard

Present:

- Goal-wise probabilities
- SIP allocation
- Risk vs return summary

Deliverables

1. Python Code
 - Monte Carlo simulation
 - Optimization logic
 - Sensitivity analysis
2. Report (2–3 pages)

Include:

- Visualization 1: Goal-wise probability comparison
- Visualization 2: Tenure vs probability
- Visualization 3: SIP vs probability
- Visualization 4: Portfolio growth bands

Reflection

Explain:

- Why probability-based planning is preferred over return-based planning
- Trade-offs between risk, time horizon, and contributions

Grading Rubric

Mathematical and Simulation Accuracy — 30%

Correct implementation of Monte Carlo simulation

Optimization Design — 25%

Proper SIP allocation and constraint handling

Sensitivity Analysis — 25%

Clarity and depth of insights

Presentation and Clarity — 20%

Quality of plots and explanation

Bonus (Optional)

Implement loan recommendation logic:

If goal probability falls below a threshold, estimate the additional lump sum or loan required to achieve the goal.